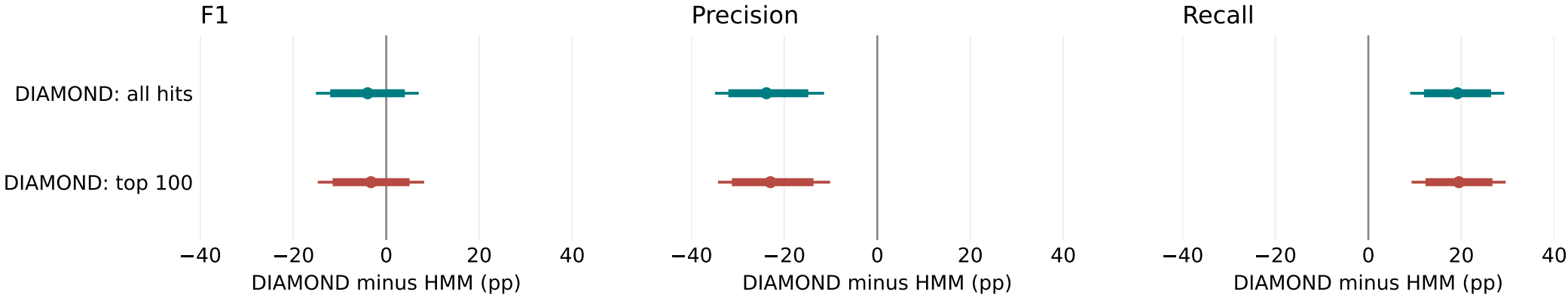


OrthoBench: initial-search replacement control

70 reference families | all 251,378 proteins retained | fixed downstream grouping and refinement

Profile expansion, candidate expansion and phylogenetic reconciliation are off in every arm.

Initial search	F1 (%)	Precision (%)	Recall (%)
HMM search	69.76	78.87	62.54
DIAMOND: all hits	65.76	55.04	81.68
DIAMOND: top 100	66.48	55.90	82.02



Thick: paired 95% CI. Thin: Bonferroni CI over six endpoints. 20,000 RefOG draws; seed 20260918.

DIAMOND increases recall and reduces precision. Both adjusted F1 intervals include zero.

Development-exposed control; search sensitivity and score calibration are not matched. No established HMM F1 advantage.